

Formation Engine — Methodology Summary

Version 3.0.0 | 13 Sections | 35 Validation Tests

Problem & Scope (§1)

The Formation Engine is a classical atomistic simulation instrument that generates molecular and crystalline structures from elemental identity and thermodynamic boundary conditions. It targets the intermediate regime (100–10,000 atoms) where quantum methods are too expensive and continuum models assume structure rather than generating it.

Design axioms: explicit units everywhere (Å, fs, kcal/mol, amu); no hidden normalization; no silent model switching; deterministic core with stochastic exploration controlled by seed.

Sole data source: the periodic table (Z=1–102). LJ parameters from the Universal Force Field (UFF). No molecular databases.

Domain: atoms with $Z \geq 6$, temperatures $T > 50$ K, system sizes $N = 2$ –10,000. Excludes excited states, quantum tunnelling, and reaction kinetics.

State Ontology (§0, §2)

The canonical state is:

$$S = (N, \mathbf{M}, \boldsymbol{\tau}, \mathbf{X}, \mathbf{V}, \mathbf{Q}, \mathbf{B}, \mathbf{F}, \mathcal{E}, \mathcal{L}, \mathcal{B})$$

partitioned into **Identity** ($N, \mathbf{M}, \boldsymbol{\tau}$ — immutable), **Phase** ($\mathbf{X}, \mathbf{V}, \mathbf{Q}, \mathbf{B}$ — evolve), and **Scratch** ($\mathbf{F}, \mathcal{E}, \mathcal{L}, \mathcal{B}$ — recomputed).

Per-particle identity vector (§0): $\mathbf{I}_i = (Z_i, A_i, Q_i, \Sigma_i, \Lambda_i, \Theta_i)$.

Energy ledger: $\mathcal{E} = (U_{\text{bond}}, U_{\text{angle}}, U_{\text{torsion}}, U_{\text{vdW}}, U_{\text{Coul}}, U_{\text{ext}})$

File hierarchy: `.xyz` (geometry) $\rightarrow$ `.xyza` (trajectory) $\rightarrow$ `.xyzc` (checkpoint with velocities + thermodynamic state + SHA-256 provenance hash).

Interaction Model (§3)

Nonbonded: Lennard-Jones 12-6 $U_{\text{LJ}} = 4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$ with Lorentz-Berthelot combining rules and quintic switching $r_{\text{on}}=9, r_{\text{cut}}=10$ Å. Coulomb $U_{\text{C}} = k_e q_i q_j / r$ with $k_e = 332.0636$ kcal·Å/(mol·e²).

Bonded: Harmonic bonds $k_b(r-r_0)^2$, harmonic angles $k_\theta(\theta-\theta_0)^2$, periodic torsions $V_n[1+\cos(n\phi-\gamma)]$, improper torsions. Parameters from UFF.

All forces computed as $\mathbf{F}_i = -\nabla_{\mathbf{x}_i} U$ through a pure-function interface: same state always yields same forces.

Thermodynamics & Units (§4)

Unit system: Å, fs, kcal/mol, amu, K, e .

$k_B = 0.001987204$ kcal/(mol·K)

$C_{\text{KE}} = 2390.057$ (amu·Å²/fs² $\rightarrow$ kcal/mol)

$k_e = 332.0636$ kcal·Å/(mol·e²)

Temperature from equipartition: $T = 2K/(N_{\text{df}} k_B)$,

$N_{\text{df}} = 3N - 3$.

Integration (§5)

Velocity Verlet (NVE): Symplectic, 2nd-order, $\Delta t = 1$ fs default. Energy drift $< 3 \times 10^{-5}$ kcal/mol/atom over 10^4 steps.

Langevin (NVT): Euler–Maruyama discretisation with friction γ and random force $\sqrt{2\gamma m_i k_B T} \xi_i$. Temperature error $< 0.6\%$.

FIRE minimisation: Damped MD with adaptive Δt . Convergence criterion $F_{\text{max}} < 0.01$ kcal/(mol·Å).

Formation Physics (§6)

A **formation** is a local PES minimum found via: (1) VSEPR geometry prediction, (2) MD exploration at elevated T , (3) periodic FIRE quenching, (4) scoring and classification.

Bond inference uses covalent radii with tolerance factor $f = 1.2$. Bonds are mutable caches, never serialised as truth.

Statistics & Scoring (§7)

Welford’s algorithm for numerically stable online mean/variance. Stationarity gate: 10 consecutive passes of coefficient-of-variation $< \epsilon$ triggers convergence. Kabsch alignment (SVD) for RMSD comparison. Multiplicative 6-factor scoring function for ranking formations.

Reaction & Electronic (§8–9)

QEq charge equilibration: minimise $E = \sum_i (\chi_i q_i + \eta_i q_i^2) + \sum_{i<j} k_e q_i q_j / r_{ij}$ subject to $\sum q_i = Q_{\text{total}}$. Fukui functions, HSAB matching, Bell–Evans–Polanyi barrier estimates. Interface defined; full MD integration pending.

Multiscale (§10)

Four regimes: molecular (2–50), cluster (50–500), bulk (500–10k, PBC + supercell), coarse-grained (10k+, CG placeholder). Supercell: $\mathbf{r}_{i,pqr} = \mathbf{r}_i + p\mathbf{a} + q\mathbf{b} + r\mathbf{c}$; bonds re-inferred post-replication.

Self-Audit (§11)

Determinism contract: same (formula, seed, params) $\Rightarrow$ bit-identical output. Enforced via seeded RNG, indexed force evaluation, IEEE 754 compliance, no fast-math.

Three Python tools: **Failure Classifier** (6 categories: NUMERICAL, PHYSICS, OOD, CONVERGENCE, TIME-OUT, UNKNOWN); **Gap Targeter** (6×10×10 grid, 95% coverage target); **Regression Detector** (4 invariants, binary pass/fail).

Validation (§12)

35 hierarchical tests across 5 levels:

Level	Tests	Status
0 — Unit system	12	100%
1 — Force evaluation	8	100%
2 — Integration	5	100%
3 — Thermodynamics	8	87%
4 — Reproducibility	3	100%

Production certified for LJ-dominated systems (noble gases, hydrocarbons, small organics). Known limitation: Coulomb–integrator coupling instability for ionic MD.

Future Work (§13)

Priority: neighbour lists ($50\times$ speedup) $\rightarrow$ NPT barostat $\rightarrow$ reactive bond orders $\rightarrow$ quantum corrections (delta-learning) $\rightarrow$ ML potentials (SchNet/MACE).

Full methodology: 11 LaTeX source files in `docs/`. Compile with `pdflatex`. MIT License.